Venkata Lakshmi Parimala Pasupuleti

Senior Full Stack AI Engineer · Python · TypeScript · React · LangGraph · RAG · FastAPI · AWS · Agentic AI · Open Source · 9yr Production

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PROFESSIONAL SUMMARY

Full stack AI engineer with 9+ years of production experience building AI-native applications across the complete stack - Python backends, TypeScript/React frontends, LLM orchestration, RAG infrastructure, and event-driven distributed systems. I ship AI systems from concept to production: at Magizhs, I took an ambiguous customer requirement and delivered a production agentic AI system (LangGraph, FastAPI, Python) with RAG over operational documents, tool orchestration, and validation guardrails in weeks as sole technical owner. AgentProbe is the open-source proof point: a multi-provider LLM evaluation platform with 3-agent LangGraph orchestration, ChromaDB vector database RAG (chunking strategies, embedding optimization), 5-metric observability engine, CI-integrated regression gate, and a 15-endpoint FastAPI backend - the kind of reusable AI infrastructure a product team contributes back to itself after each new workflow. LeaseI is the production AI application: an 8-agent LangGraph system with conditional routing, long-term memory, RAG over multi-source documents, and deterministic HITL safety gates - exactly the orchestration and knowledge retrieval architecture Kaya builds for mission-critical workflows. At Entrata, I owned production enterprise systems at scale: RabbitMQ event-driven async architecture, AWS S3/EC2/RDS/Glue, Docker microservices, PostgreSQL, Redis - 99.5% uptime, zero security incidents, 9yr. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Full Stack & Languages: Python (9yr), TypeScript, JavaScript, React.js, FastAPI, Flask, REST APIs, Node.js, HTML/CSS, SQL, Bash, Git, Linux

AI Orchestration & RAG: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM Orchestration, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Chunking Strategies, Semantic Search, LLM Evaluation, Hallucination Detection, HITL, Prompt Engineering, MCP, Anthropic API, OpenAI API, Ollama, Claude Code, NLP, BERT, spaCy, PyTorch, scikit-learn

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (event-driven async), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Distributed Systems, Performance Engineering, PHI/HIPAA Compliance, Agile/Scrum

PROJECTS

AgentProbe: Open-Source AI Orchestration & RAG Platform

LLM Orchestration · RAG · Vector DB · Open Source · CI/CD

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama · MCP | github.com/venkatalakshmiparimala/AgentProbe

Open-source full stack AI platform - the reusable orchestration, RAG, and observability infrastructure that underpins production AI workflows. **LLM orchestration:** 3-agent LangGraph pipeline (Orchestrator, RAG Tracer, Auto Fix) with multi-provider routing (Anthropic, OpenAI, Ollama) via clean API abstractions - swappable without prompt rewrites. **RAG infrastructure:** ChromaDB vector database, chunking strategy optimization, embedding selection, semantic search. **Observability:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); CI-integrated regression gate (GitHub Actions) blocking quality degradation pre-deployment. **Backend:** FastAPI with 15 production endpoints. Zero regressions across 1,240+ test cases. Enterprise-deployable, fully documented.

LeaseI: 8-Agent Production AI - Orchestration, Memory & HITL

Multi-Agent · RAG · Long-Term Memory · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Full stack production AI application - the agent orchestration and knowledge retrieval architecture for mission-critical workflows where decisions have real-world consequences. **Multi-agent orchestration:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with real conditional routing and dynamic task delegation. **Knowledge systems:** OCR document ingestion, section-aware chunking, TF-IDF retrieval with cosine reranking; shared SQLite knowledge repository across all agents - documents, settings, historical incidents, and audit logs persisted across sessions. **HITL safety gate:** deterministic escalation when confidence below threshold; 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Full Stack AI Application - Enterprise API Integration

Full Stack · TypeScript · React · REST APIs · Docker

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full stack AI application integrating LLMs with enterprise REST APIs - Python/Flask backend with NLP-powered query translation; React.js + TypeScript frontend; Docker containerization; HIPAA-compliant RBAC with authentication and full data isolation; ~60% reduction in manual retrieval time. Demonstrates the full stack AI integration pattern: consuming enterprise operational data APIs and exposing them through AI-powered interfaces that non-technical users can operate directly.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Zero-to-Production Full Stack AI Delivery

- Took ambiguous customer requirements and delivered a production agentic AI system (LangGraph, FastAPI, Python) end-to-end in weeks - gathered requirements directly from stakeholders, designed multi-step reasoning pipeline with RAG over operational documents, tool orchestration, and HITL review gates; sole technical owner from architecture through production deployment and documentation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems; translated complex computational outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Distributed systems and event-driven infrastructure** - RabbitMQ async processing system for gross-rent updates across thousands of tenant records (retry logic, idempotency guarantees, dead-letter handling); AWS S3/EC2/RDS/Glue production data pipelines; Redis caching; Docker microservices (50% scalability improvement, 30% deployment time reduction); 99.5% uptime across multi-tenant SaaS.
- **Full stack production delivery** - 20+ full-stack modules (PHP, Python, PostgreSQL, JavaScript/TypeScript, React); 200+ SQL optimizations (30% performance gain); HUD compliance pipeline (180,000 domain records, 100K+ housing units); Python/Bash automation eliminating 80% of manual operational workload; GitHub Actions CI/CD; 100% PHI compliance, zero security incidents across 7yr.
- **Technical leadership** - 120+ Agile sprints; 300+ peer code reviews; internal debugging tools reducing triage time 40%; mentored 5+ engineers; partnered directly with product and operations stakeholders to define and ship new capabilities; leveraged Claude Code to accelerate engineering delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017